Testing Guide

Aegivex AI

Version	1.0
Document Type	Testing Guide
Project	OKX.AI Genesis Hackathon
Prepared For	Development Team
Prepared By	Aegivex AI Team

1. Overview

This document defines the testing strategy for Aegivex AI.
The objective is to ensure that every feature functions correctly, the platform remains secure, AI responses are reliable, and the application is stable before deployment.

2. Testing Objectives

The testing process should verify:

- Functional correctness
- Security
- Performance
- Reliability
- AI response quality
- User experience
- API stability

3. Testing Types

The project includes:

- Unit Testing
- Integration Testing
- API Testing
- UI Testing

- AI Response Testing
- Database Testing
- Security Testing
- Performance Testing
- User Acceptance Testing (UAT)

4. Unit Testing

Purpose: Verify individual functions and modules.
Examples:

- Wallet validation
- Risk score calculation
- Authentication logic
- Input validation
- Utility functions

Expected Result: Every function should return the expected output for valid and invalid inputs.

5. Integration Testing

Purpose: Verify communication between system components.
Test Cases:

- Frontend → Backend
- Backend → Database
- Backend → AI Engine
- Backend → External APIs

Expected Result: All services communicate successfully without data loss or unexpected failures.

6. API Testing

Verify every REST API endpoint.
Examples:

- User Registration
- Login
- AI Chat
- Wallet Scanner
- Token Scanner
- Smart Contract Scanner
- Website Scanner
- Transaction Explainer

- Dashboard
- Scan History

Validation Areas:

- HTTP Status Codes
- JSON Response Format
- Authentication
- Error Handling

7. UI Testing

Verify all user interface components.

Checklist:

- Navigation
- Buttons
- Forms
- Cards
- Dashboard
- Mobile Responsiveness
- Loading States
- Error Messages

Expected Result: The interface should be responsive, intuitive, and free of visual issues.

8. AI Response Testing

Verify AI output quality.

Checklist:

- Correct understanding of prompts
- Accurate blockchain explanations
- Consistent risk scoring
- Clear recommendations
- No fabricated information

Expected Result: AI responses should be accurate, understandable, and actionable.

9. Database Testing

Verify database operations.

Checklist:

- Insert operations
- Update operations
- Delete operations

- Read operations
- Foreign key relationships
- Data consistency
- Index performance

10. Security Testing

Verify platform security.

Testing Scope:

- JWT Authentication
- Authorization
- SQL Injection Protection
- XSS Protection
- Input Validation
- API Rate Limiting
- Secure Password Storage
- HTTPS Configuration

11. Performance Testing

Measure application performance.

Metric / Target Area	Target Threshold
API Response Time	< 500 ms (excluding AI processing)
AI Response Time	< 5 seconds
Dashboard Loading	< 2 seconds
Concurrency	Support concurrent users

Monitoring Parameters:

- CPU Usage
- Memory Usage
- Database Performance
- API Latency

12. User Acceptance Testing (UAT)

Users should verify:

- Registration
- Login
- Wallet Scan
- Token Scan
- Contract Scan
- Website Scan
- Transaction Explanation
- AI Chat
- Dashboard
- Scan History

Expected Result: Users can complete all core tasks successfully.

13. Error Handling Tests

Verify the application correctly handles:

- Invalid wallet addresses
- Invalid contract addresses
- Invalid URLs
- Invalid transaction hashes
- Empty inputs
- Expired JWT tokens
- Database failures
- AI service failures
- Network interruptions

14. Regression Testing

Before every release:

- Re-test all core features
- Verify previous bugs remain fixed
- Confirm no new issues were introduced

15. Test Checklist

Before deployment ensure:

- All APIs pass testing

- Database operations succeed
- Authentication works correctly
- AI responses are validated
- UI is responsive
- No critical security issues exist
- Performance targets are achieved
- No critical bugs remain

16. Expected Outcome

The testing process ensures that Aegivex AI is stable, secure, reliable, and ready for production use. Every feature should perform as expected, AI responses should remain accurate, and users should experience a fast and seamless Web3 security platform.